Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 · Energy storage devices, hydrogen energy, and sensors · Supercapacitors , Ragone plot

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Supercapacitors , Ragone plot

1 Supercapacitors , Ragone plot

Study focus: Supercapacitors , Ragone plot. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000134 · EV0000135

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module five on energy storage devices on supercapacitors and this will be the last class of this module.

Key details

- When we use a power source, the negative terminal is attached to one of the plates so that gets the negative charge and the positive terminal when it is attached to the other plate that plate gets the positive charge.
- The charge stored in a capacitor is directly proportional to the voltage developed across the capacitor.
- Then C is known as the capacitance and the unit used is ferrats.
- will decide the capacitance of the capacitance because capacitance is directly proportional to the area of the plate as you can see in the picture here.
- This is the area of the plate and it is inversely proportional to the distance between the two plates.
- What are supercapacitors then?
- They are also known as ultra capacitors or electrochemical double layer capacitors, E, D, and C.
- you can understand that the capacitance has gone up to such a high extent and they also charge and discharge very quickly.
- This is for any capacitor and of course it is applicable to supercapacitor also that they charge and discharge very quickly.
- Why is the capacity so much higher than that of a capacitor?
- We have electrodes, we have the electrolyte and we have a separator.
- Very important high surface area materials.

- The separator is an iron permeable separator and it's.
- As in batteries, it is placed between the electrodes, and what is the role of a separator?
- It is always to prevent electrical contact between the anode and the cathode, but it is going to allow the ions to.

Example or application from the lecture

- Before we talk about supercapacitors, let us understand what is a capacitor and how is a supercapacitor different from a capacitor.
- If we know that capacitor is usually in the capacitance is given in microfarads, but for a supercapacitor it will be listed in thousands of ferrats.
- Let us look at the construction.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 05:00–09:59

move from the one electrode to the electrode to the other electrode and the example for this would be porous polypropylene.

Key details

- How exactly are these components placed?
- You can see in this picture.
- The separator is sandwiched between the electrodes.
- The high surface area electrode which are highly porous.
- And then this whole system is now placed in a case and then electrolyte is poured over it.
- It goes through all this porous electrodes which we have placed over here.
- And then of course the electrodes we need.
- Just like in a battery, we need current collectors.
- When a potential is applied, so the positive electrode, one of the electrode obviously will become positive based on the terminal of the external source.
- And similarly, the negative electrode which is connected to the negative terminal is going to become is going to get attract the positive ions.
- So and it and this double layer now because you made a porous electrode it is formed at the entire electrode electrolyte interface and plus the charge separation is just nanometer range.
- What is happening here?
- Only absorption and desorption.
- They get they form the double layer and then when you are discharging then these ions are getting away from the electrode surface.
- that is happening here.
- So because of this.
- We find that this supercapacitor has many advantages, which we will discuss now.
- It is just this absorption and desorption which happens during the charging and the disorganization.

- charging process now what has happened the distance between the charge lay has become into the nano range as I explained just now so D is very small and.
- Also we used porous electrodes.
- So because of the porous electrode, what do we get?
- We have a very high surface area.
- The moment you make a material porous, we know that it is going to get in get a very high surface area.

Example or application from the lecture

- Let us look at the working.

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Core explanation · 10:00–14:59

We know about capacitance C is proportional to A by D.

Key details

- If the area is very high and the D is very small, in fact here it's in the nanometer scale.
- Now this formation of electric double layer of course takes place at both the electrode electrolyte interface and therefore at each of these electrodes we have the capacitance.
- And the total capacitance would be 1 by C T is equal to 1 over C plus 1 over C minus.
- What are the advantages of these supercapacitors?
- They charge very, very quickly.
- Because it just involves the movement of these ions towards the electrodes and forming of the double layer.
- And very high.
- power density because power density depends on how fast the energy is getting discharged and we know that they can discharge also very quickly because there is no redox reaction that is taking place.
- It's only desorption of ions that is required.
- And of course high cycle life they can be cycled millions of times as compared to your battery, which the best batteries have about five hundred to thousand is very high cycle life, but here we have device which can be cycled several time because again there is no redox reaction that has to take place.
- It is just the absorption and desorption of ions which can happen as many times like we can say it's almost unlimited cycle life.
- And yes, compared to batteries, they are much safer as the internal resistance is very low, and when the internal resistance is low, we know that the heating rate is also going to be low.
- Then there are certain disadvantages also.
- What is it that they have very low energy density.
- Low energy density.

- What is the reason that it the reason is they discharge very quickly, so obviously they cannot retain or hold the charge for a long time.
- So energy density is low, and of course, high self-dcharge because it just involves absorption and disorption of ions.
- High self discharge.
- Linear discharge voltage.
- If you remember for a battery when we draw voltage versus time, the line looks like this the graph looks like this what happens it remains steady for a long time and then there is a drop in the voltage when we say the battery is dead.
- Whereas for a supercapacitor, the drop is linear.
- So therefore the volta is going to drop very quickly with time, whereas in a battery we find that we can get a steady voltage for a reasonable amount of time.
- This supercapacitor because of the electrodes which are you being used, very highly porous electrodes have to be used.
- They are very expensive.
- And then the power is available for a very short duration because they discharge very quickly.
- Where would such.
- Batteries be useful or supercapacitors be useful.
- They are used for memory backup.
- Then hybrid cars for start stop applications in flash photography where you need high power for a very short period of time.
- flashlights, portable media players, and also as an intermediate energy storage for FM radios, cell phones and emergency kits.
- We have talked about different types of batteries, we talked about fuel cells, we talked about supercapacitor.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 15:00–19:59

Energy storage devices comparison compare the performance of these and that can be done when we plot energy density.

Key details

- Energy density versus power density for a whole range of energy storage devices and why is it possible to use to plot all these energy different energy storage devices?
- That's because it's a double log chart.
- The reason that they discharge very quickly, so they can give you a very high power density.
- They have very high power density, but they don't retain the charge for a long time, hence they have low energy density.
- The reason being that there is no redox reaction.
- is simple absorption and desorption of ions that has to take place and therefore they charge as well as discharge very quickly.

- We also studied about fuel cells now fuel cells have very high energy density but low spow density.
- The reactions have to take place, and the reactions are usually very slow.
- If you remember, we always use catalysts.
- So therefore the reactions are slow, and hence the few if our fuel cells the power density is going to be very low, whereas the energy density is high because we can supply the fuel and the energy and it can work as long as you supply the fuel so the energy density is very high.
- Batteries of course come somewhere in between they are occupying this position, solid acid battery, nickel cadmium battery, lithium ion battery, lithium ion battery.
- Has an advantage because of lithium being used, which is a lightweight metal.
- And of course, when you talk of combustion engines, we know that we can keep on supplying the gasoline and therefore the energy density is going to be high and the reactions are very quick where we know that chain reactions happen.
- So therefore the power density is also very high.
- So fuel cells will have high energy density and whereas they have low power density batteries will have moderate energy density and power density ultra capacitors or supercapacitors they can deliver very high power therefore power density is very high whereas the.
- Energy density is going to be low for a ultra capacitor or a supercapacitor.
- So in the this is the end of this module, and in the next class, we are going to talk about.
- Then next topic and that is corrosion.

Example or application from the lecture

- that is given in watt hour per kg versus the power density that is watt per kg of a particular device and such a plot is known as a raguni plot.
- Let us have a look at this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Supercapacitors , Ragone plot transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

C Revision and retrieval practice